Chapter 6: Haloalkanes & Haloarenes

Named Reactions Sheet + Tips & Tricks

Formula Sheet

I. Preparation of Haloalkanes

Free Radical Halogenation of Alkanes

$$CH_4 \xrightarrow[h\nu \text{ or } 520\text{-}670\text{ K}]{Cl_2} CH_3Cl + CH_2Cl_2 + CHCl_3 + CCl_4$$

Not preferred in the lab since it does not stop at monosubstitution; used industrially.

Alkane + Iodine (Reversible)

$$CH_4 + I_2 \overset{\text{Heat}}{\rightleftharpoons} CH_3I + HI$$

HI is a strong reducing agent and reduces CH_3I back to CH_4; a strong oxidising agent (HIO_3 or HNO_3) is added to oxidise HI and stop the reaction: $5\,HI + HIO_3 \longrightarrow 3\,I_2 + 3\,H_2O$.

Alkene + Hydrogen Halide (Markovnikov Addition)

$$CH_2{=}CH_2 + HBr \longrightarrow CH_3{-}CH_2{-}Br$$

Markovnikov's Rule: H attaches to the carbon already bearing more H atoms; general form: $R\text{-}CH{=}CH_2 + HX \rightarrow R\text{-}CH(X)\text{-}CH_3$.

Anti-Markovnikov (Peroxide/Kharasch Effect)

$$R{-}CH{=}CH_2 + HBr \xrightarrow{(C_6H_5COO)_2O_2,\text{ peroxide}} R{-}CH_2{-}CH_2Br$$

Works only with HBr (not HCl or HI); free-radical mechanism reverses the addition orientation.

Allylic Substitution – NBS/SO2Cl2

$$CH_3{-}CH{=}CH_2 \xrightarrow[400\text{-}500\,K]{Cl_2 \text{ or } SO_2Cl_2} CH_2{=}CH{-}CH_2{-}Cl$$

NBS (N-bromosuccinimide) gives allylic substitution by Br; SO_2Cl_2 (sulfuryl chloride) gives allylic substitution by Cl.

From Alcohols – Lucas Reagent

$$R{-}OH + HCl \xrightarrow{\text{anhy. } ZnCl_2} R{-}Cl + H_2O$$

Conc. HCl + anhydrous $ZnCl_2$ = Lucas reagent. Mechanism: $ZnCl_2$ (Lewis acid) accepts a lone pair from alcohol O, activating OH as a leaving group; R^+ carbocation forms, then Cl^- attacks. Rate: $3° > 2° > 1°$ alcohols – basis of the Lucas test.

From Alcohols – PCl_5

$$R{-}OH + PCl_5 \longrightarrow R{-}Cl + POCl_3 + HCl$$

PCl_5 is a solid that sublimes to gas on heating.

From Alcohols – PCl_3

$$3\,R{-}OH + PCl_3 \longrightarrow 3\,R{-}Cl + H_3PO_3$$

From Alcohols – $SOCl_2$ (Darzens Process)

$$R{-}OH + SOCl_2 \longrightarrow R{-}Cl + SO_2 + HCl$$

Considered the best method: both by-products (SO_2, HCl) are gaseous, leaving a clean purified R-Cl.

Halogen Exchange – Finkelstein Reaction

$$R{-}X + NaI \xrightarrow{\text{Acetone}} R{-}I + NaX$$

Exclusive for alkyl iodides (X = Cl or Br). NaI is soluble in acetone; NaCl/NaBr precipitate out, driving the reaction forward.

Halogen Exchange – Swarts Reaction

$$R{-}X + AgF/Hg_2F_2/SbF_3 \longrightarrow R{-}F$$

Exclusive for alkyl fluorides.

Hunsdiecker Reaction (Borodine Reaction)

$$R{-}COOH \xrightarrow[-H_2O]{AgOH} R{-}COOAg \xrightarrow[CCl_4]{Br_2} R{-}Br + CO_2 + AgBr$$

Exclusive for alkyl bromides; carboxylic acid → silver salt → treated with Br_2 in CCl_4, silver carboxylate loses CO_2.

II. Reactions of Haloalkanes – Nucleophilic Substitution

Substitution by -OH (Formation of Alcohol)

$$C_2H_5-Br \xrightarrow{aq \cdot NaOH} C_2H_5-OH + NaBr$$

S_N2 mechanism; no carbocation formed.

Substitution by -OR (Williamson's Synthesis)

$$CH_3Cl + C_2H_5ONa \xrightarrow{ether} CH_3-O-C_2H_5 + NaCl$$

Forms ether (here: sodium ethoxide); alkyl halide must be 1°.

Substitution by -CN (Alkyl Cyanide/Nitrile)

$$C_2H_5-Br + KCN \xrightarrow{alc \cdot} C_2H_5CN + KBr$$

Attack occurs via carbon of CN^- (ionic bond); product has one more carbon than R-X.

Substitution by -NC (Alkyl Isocyanide)

$$C_2H_5Br + AgCN \longrightarrow C_2H_5NC + AgBr$$

Ag-CN is covalent (Ag^+: $5s^0 4d^{10}$), so N attacks instead of C, forming isocyanide – follows Fajan's Rule.

Substitution by Nitrite – via Oxygen (Alkyl Nitrite/Ester)

$$C_2H_5Br + KONO \longrightarrow C_2H_5-O-N{=}O + KBr$$

KNO_2 (ionic) attacks via oxygen, forming the nitrite ester.

Substitution by Nitro Group – via Nitrogen (Nitroalkane)

$$C_2H_5Br + AgNO_2 \longrightarrow C_2H_5NO_2 + AgBr$$

$AgNO_2$ (covalent) attacks via nitrogen, forming the nitroalkane.

Substitution by -NH_2 (Amine Formation)

$$R-X + NH_3 \longrightarrow R-NH_2 + HX$$

Excess NH_3 prevents over-alkylation; excess R-X drives further substitution to 2°, 3° amine, and finally quaternary ammonium salt.

III. Elimination and Reaction with Metals

Elimination Reaction (Saytzeff Rule / Beta-Elimination)

$$CH_3-CH_2-CH(Br)-CH_3 \xrightarrow{alc\cdot KOH} CH_3-CH=CH-CH_3 + HBr$$

Alcoholic KOH removes H from the beta carbon, forming the more substituted (major) alkene. Reagent contrast: aqueous KOH gives substitution (alcohol); alcoholic KOH gives elimination (alkene).

Wurtz Reaction

$$2\,R-X + 2\,Na \xrightarrow{\text{dry ether}} R-R + 2\,NaX$$

Couples two alkyl groups; works best for identical R groups.

Frankland Reaction (with Zinc)

$$R-X + Zn + X-R \xrightarrow{\text{dry ether}} R-R + ZnX_2$$

Grignard Reagent Formation

$$R-X + Mg \xrightarrow{\text{dry ether}} R-Mg-X$$

Alkyl magnesium halide.

Organolithium Formation

$$R-X + Li \xrightarrow{\text{dry ether}} R-Li + LiX$$

Gilman Reagent

$$R-Li + CuI \longrightarrow R_2CuLi + LiI$$

Lithium dialkylcuprate.

Reduction with $LiAlH_4$

$$R-X + LiAlH_4 \longrightarrow R-H + LiX + AlH_3$$

$LiAlH_4$ acts as a source of hydride ion, displacing the halide in an S_N2 substitution – reduces alkyl halide to the parent alkane.

IV. Chloroform, Iodoform, Freons & DDT

Chloroform Preparation – Haloform Reaction

$$CH_3CH_2OH + Cl_2 \longrightarrow CH_3CHO + 2\,HCl$$

Ethanol/acetone treated with bleaching powder ($CaOCl_2$); overall sequence continues to trichloroacetaldehyde then chloroform: $Ca(OH)_2 + 2\,CCl_3CHO \longrightarrow 2\,CHCl_3 + Ca(HCOO)_2$.

Iodoform Preparation – Haloform Reaction

$$CH_3CH_2OH + 4\,I_2 + 6\,NaOH \longrightarrow CHI_3(\downarrow) + HCOONa + 5\,NaI + 5\,H_2O$$

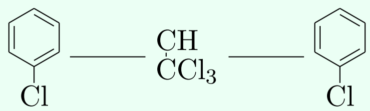
Ethanol treated with I_2 and NaOH; yellow crystalline solid with characteristic antiseptic smell.

Structure of Freons

Cl
Cl-C-F
F

Freons are chlorofluorocarbons (CFCs), general formula $C_xCl_yF_z$; shown here is CCl_2F_2 (Freon-12). Used as refrigerants and aerosol propellants; stable in the lower atmosphere but deplete the ozone layer in the upper atmosphere.

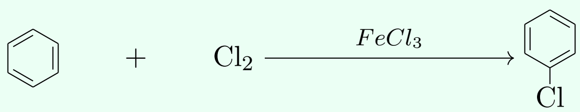
Structure of DDT



DDT (dichlorodiphenyltrichloroethane): two *p*-chlorophenyl rings attached to a central carbon bearing a -CCl_3 group. A powerful insecticide, once widely used to control insect-borne diseases; now restricted due to environmental persistence.

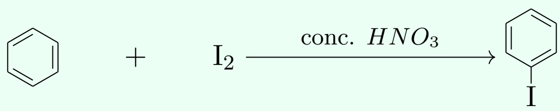
V. Haloarenes – Preparation

Direct Halogenation of Benzene



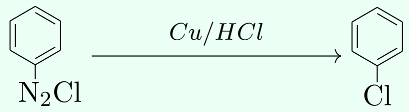
Any Lewis acid catalyst can be used: $FeCl_3$, $FeBr_3$, $AlCl_3$, $AlBr_3$, $ZnCl_2$, BF_3.

Iodination (needs oxidising agent)



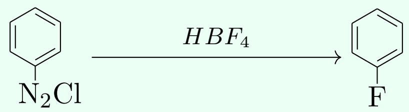
Reaction is reversible and HI (a strong reducing agent) would reduce the product back; an oxidising agent is required.

Gattermann Reaction



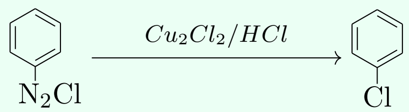
Aryl diazonium salt converted directly to aryl halide using copper powder with HCl/HBr.

Balz-Schiemann Reaction



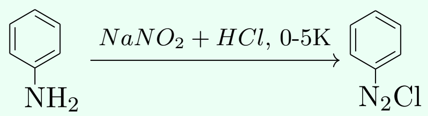
Standard route to fluorobenzene since direct fluorination does not work.

Sandmeyer Reaction



Cuprous halide catalyses the substitution (also works with Cu_2Br_2/HBr for the bromo product) – distinct from Gattermann, which uses Cu powder directly.

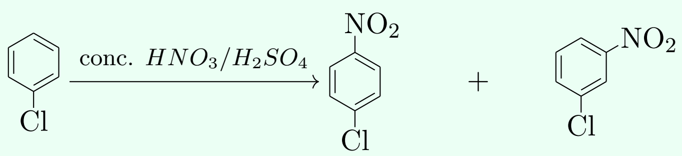
Diazotisation of Aniline



Forms benzenediazonium chloride – gateway to Sandmeyer, Gattermann, and Balz-Schiemann above.

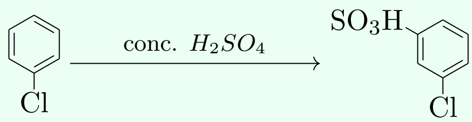
VI. Reactions of Haloarenes

Nitration



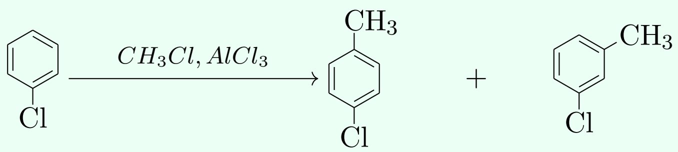
Chlorine is o, p-directing despite being a deactivating group ($-I$ dominates, but $+M$ still stabilises the arenium ion intermediate best at o, p).

Sulphonation

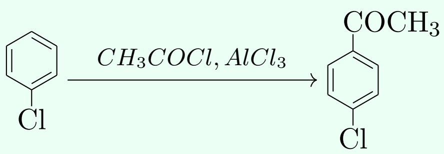


Gives predominantly the para product with fuming/conc. H_2SO_4.

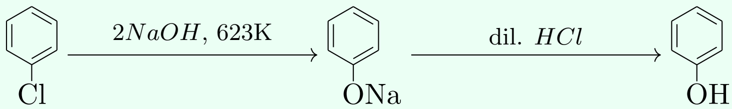
Friedel-Crafts Alkylation



Friedel-Crafts Acylation

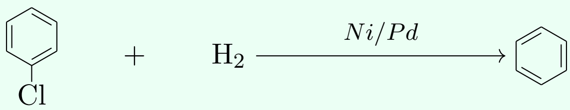


Nucleophilic Substitution – Dow's Process (to Phenol)

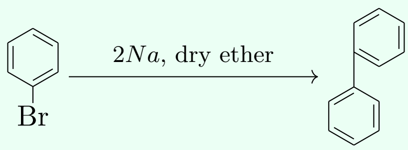


Requires drastic conditions (high temperature/pressure) since haloarenes resist normal nucleophilic substitution.

Reduction to Benzene



Fittig Reaction



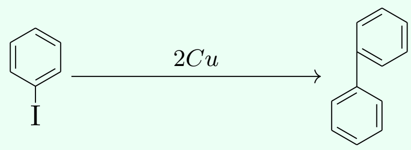
Reaction with sodium; couples two aryl halides (compare Wurtz for alkyl halides) to give biphenyl.

Wurtz-Fittig Reaction

$$C_6H_5Cl + 2\,Na + CH_3Cl \xrightarrow{\text{dry ether}} C_6H_5CH_3 + 2\,NaCl$$

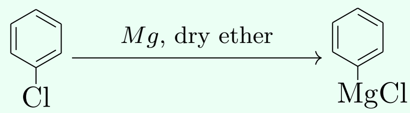
Couples an aryl halide with an alkyl halide.

Ullmann Reaction



Works for iodobenzene only, using copper.

Aryl Grignard Formation



Tips, Tricks & Hacks

- Bond length/strength trend down the halogen group: C-F shortest/strongest, C-I longest/weakest – this is why alkyl iodides are the most reactive and alkyl fluorides the least reactive toward nucleophilic substitution.
- Distinguish 1°/2°/3° alkyl halides via Lucas test turbidity time: 3° (white ppt. immediately) > 2° (5–10 min) > 1° (no turbidity at room temp).
- Carbocation stability order governs which product forms in Markovnikov addition: 3° > 2° > 1°.
- In haloarenes, the ring carbon is sp^2 hybridised (vs. sp^3 in haloalkanes) giving a shorter, stronger C-X bond; resonance delocalises the halogen's lone pair into the ring giving partial double-bond character; and the ring sterically shields the carbon from backside nucleophilic attack – three reasons haloarenes resist normal nucleophilic substitution.
- Fluorobenzene cannot be made by direct halogenation (the F-ring bond formed is too reactive) – it must be made via the Balz-Schiemann reaction.
- Chloroform must be stored in dark coloured (amber) bottles because it is slowly oxidised by air and light into the highly poisonous gas phosgene ($COCl_2$).